Towards Dynamic-aware SLAM with 4D Reconstruction

Youran Wang

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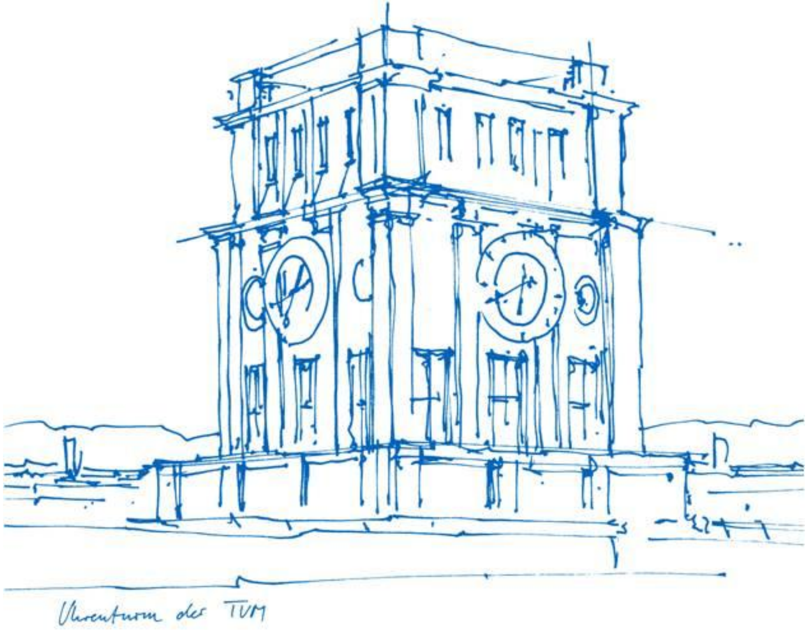

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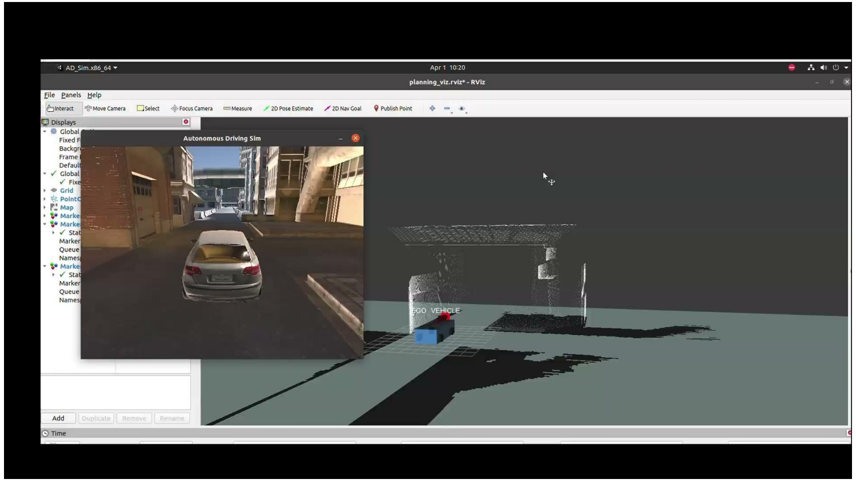
github:https://github.com/Euzil
gitlab: https://gitlab.lrz.de/go73hay

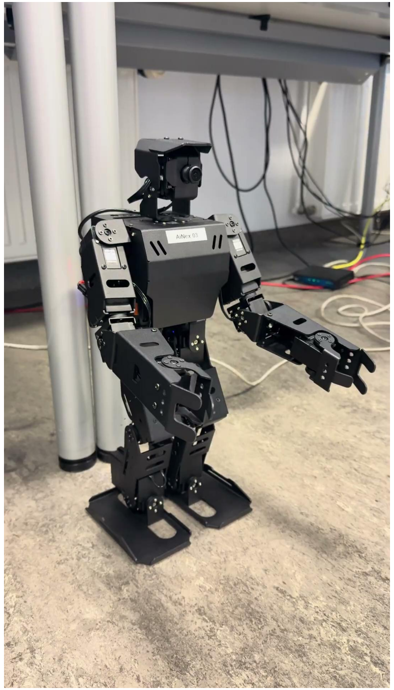
Robotics, Cognition, Intelligence
Technical University of Munich, Munich
Apr. 2024 - laufend

Robotik und Autonomie System
University of Lübeck, Luebeck
Sept. 2019 - März 2024

My Projekts before:

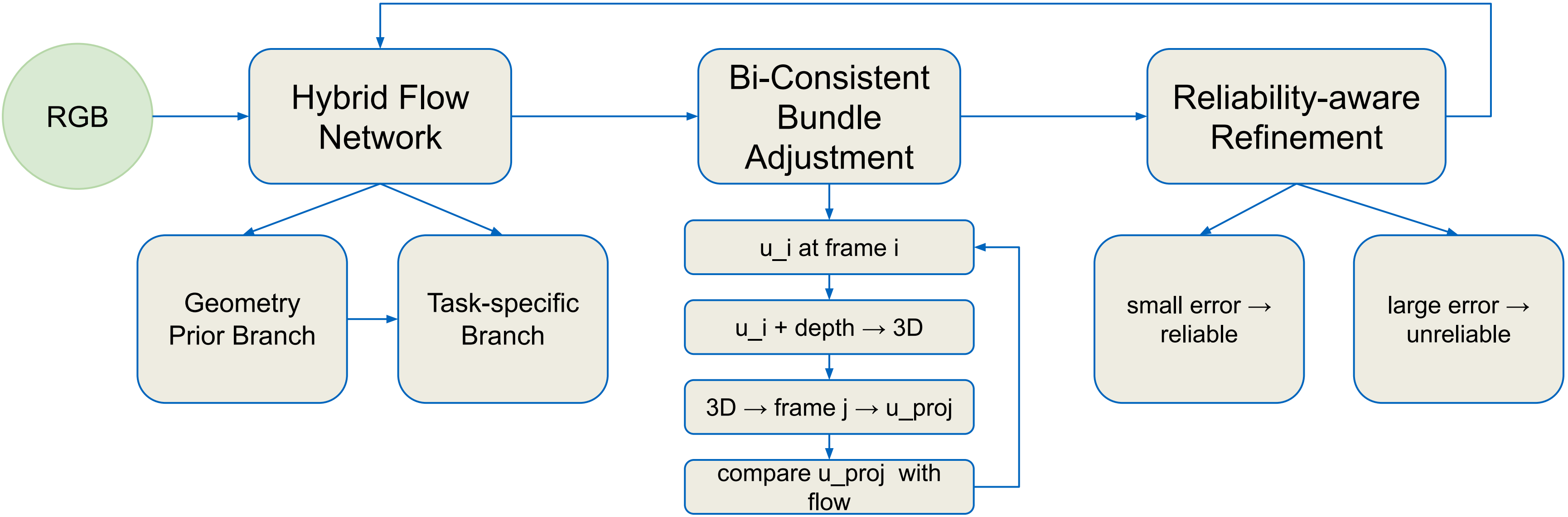
Autonomous Driving Project
https://github.com/Euzil/Introduction_to_ROS
Sim2Real Challenge 2025
https://github.com/Euzil/SIM2REAL-2025
Cristiano Roboto
https://github.com/Euzil/Christiano_Roboto





Review: FoundationSLAM: Unleashing the Power of Depth Foundation Models for End-to-End Dense Visual SLAM

- matching without Geometry
- no multi-view consistency



Review: V-DPM: 4D Video Reconstruction with Dynamic Point Maps

DUSt3R / VGGT

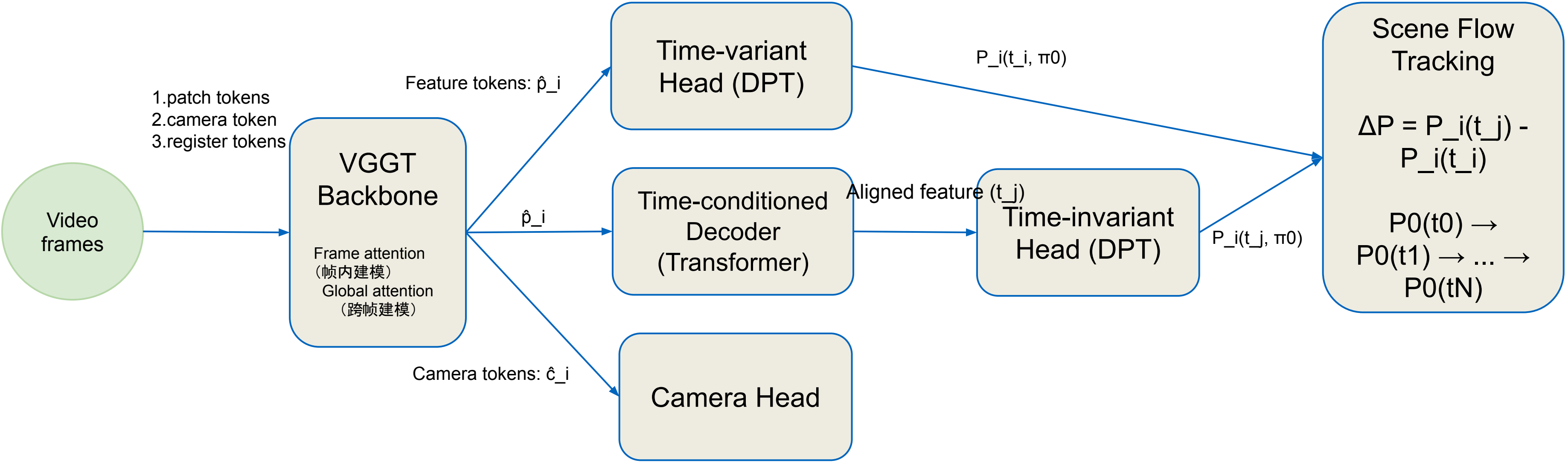
- Only static scenes can be processed

MonST3R

- It can perform 4D operations
- relies on 2D tracking (optical flow/keypoints) and is not a "pure 3D representation

Dynamic Point Maps（DPM）

- A unified representation of 3D + motion
- only handle two images (pairwise); multi-frame processing requires optimization (slow and complex)



FoundationSLAM

conclusion

- More accurate (lower ATE)
- Reconstruction is better (lower Chamfer distance).
- Real-time (18 FPS)

Innovation

1. Deep fusion of Flow and Geometry;
2. Bi-consistent BA;
3. Residual-driven structured correction;
4. True closed-loop optimization (end-to-end).

V-DPM

conclusion

- Compared to previous methods:
  error rate ↓ ~5 times

- In multi-frame scenarios: DPM performance degrades
  V-DPM remains stable

- Qualitative
  Smoother, More consistent, No more breakage

Innovation

1. Two-stage decomposition;
2. Time-conditioned transformer;
3. Transfer learning using static models.

Q&A